A Media-Optimized Iterative Detection Scheme for Dual-Layer Staggered BPMR Systems

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This paper proposes a media-optimized iterative detection scheme for a dual-layer staggered bit-patterned magnetic recording (DL-BPMR) system operating at a combined areal density of 4.0 Terabit per square inch (Tb/in^2), 2.0 Tb/in^2 per layer. The bit-island diameters of both recording layers are optimized to mitigate three-dimensional interference, including inter-symbol, inter-track, and interlayer interferences. In addition, a 0.25T_x oversampling scheme (T_x is a bit period) combined with a soft-information adjustment technique is introduced to improve the reliability of the log-likelihood ratio during iterative detection. Simulation results demonstrate that the proposed system achieves superior bit-error rate performance compared to conventional DL-BPMR and single-layer BPMR systems.

***Index Terms*— Bit-patterned magnetic recording, dual-layer magnetic recording, soft-information adjustment, three-dimensional interference.**

I. Introduction

The exponential growth of data generated by artificial intelligence (AI) ecosystems has imposed increasingly stringent demands on mass storage systems. Hard disk drives (HDDs) remain the predominant storage medium in data centers owing to their cost-per-capacity advantage. Nevertheless, conventional perpendicular magnetic recording (PMR) technology is approaching a fundamental physical limit at approximately an areal density (AD) of 1 Tb/in^2, due to the superparamagnetic limit [1]. To overcome this limitation, bit-patterned magnetic recording (BPMR) has been proposed as a promising next-generation recording technology [2], wherein isolated magnetic islands each store exactly one bit, enabling ADs beyond 4 Tb/in^2.

To further increase storage capacity, dual-layer bit-patterned magnetic recording (DL-BPMR) has been proposed [3], allowing data to be stored across two staggered recording layers. This configuration effectively doubles the areal density. However, the DL-BPMR technology introduces additional complexity in signal recovery. A single read head simultaneously captures superimposed readback contributions from both layers, yielding a composite signal degraded by inter-symbol interference (ISI), inter-track interference (ITI), and interlayer interference (ILI). To address these issues, this paper proposes a joint media optimization and iterative signal detection scheme for DL-BPMR systems.

II. Channel Model and Proposed System

A dual-layer BPMR medium is considered, as illustrated in Fig. 1(a). The recording medium is modeled as a two-dimensional image with a spatial resolution of 0.25×0.25 nm^2 per pixel. The upper and lower recording layers are denoted by RL2 and RL1, respectively. The bit period (T_x) and track pitch (T_z) are both set to 18 nm, corresponding to an AD of 2.0 Tb/in^2 per layer. The lower layer (RL1) is offset from the upper layer (RL2) by 0.5T_x in the down-track direction. As shown in Fig. 1(b), both layers have a thickness of 2 nm. The head-media spacing (HMS) relative to RL1 and RL2 is set to 5 nm and 2 nm, respectively [4].

Fig. 2 shows a block diagram of the proposed system. User bit sequences u_k^{Top} and u_k^{Bot} are encoded by a low-density parity-check (LDPC) encoder [5], producing coded sequences a_k^{Top} and a_k^{Bot}, which are written onto RL2 and RL1, respectively. During readback, the mixed signal is oversampled and processed by a 1-D equalizer, followed by a 1D-SOVA detector, to generate soft information in the form of log-likelihood ratios (LLRs). The LLRs are refined by the proposed SIA process prior to LDPC decoding, yielding estimated user sequences $\hat{u}_k^{\mathrm{Top}}$ and $\hat{u}_k^{\mathrm{Bot}}$.

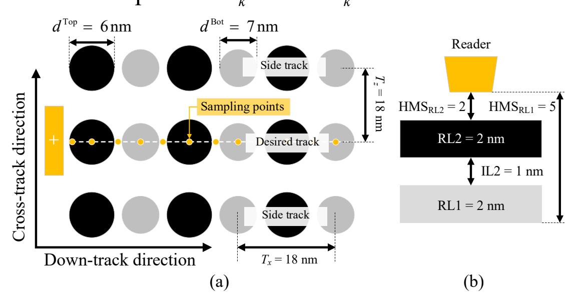

Fig. 1. Geometry of the dual-layer BPMR medium, (a) top view and (b) cross-section view under an AD of 2.0 Tb/in^2 per layer.

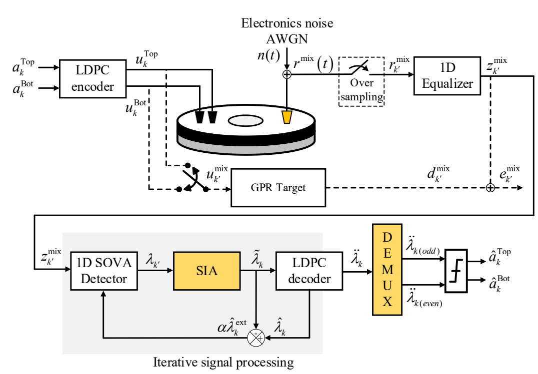

Fig. 2. Block diagram of the proposed dual-layer BPMR system with SIA.

A. Dual-Layer Medium Geometry and Bit-Island Optimization

To suppress 3D interference, the bit-island diameters of both layers are jointly optimized over a range from 6 nm to 12 nm. To directly evaluate their impact on intrinsic detection performance, optimization is conducted without LDPC decoding. The simulation results in Fig. 3 show that the optimal diameters are d^{Top} = 6 nm and d^{Bot} = 7 nm, which provide the minimum average bit-error rate (BER) across both layers. These values are therefore adopted in all subsequent simulations.

B. Oversampled Iterative Detection with SIA

Based on the $0.5T_x$ interlayer offset, a $0.25T_x$ oversampling rate is used to capture the composite readback waveform better. The oversampled sequence is defined as $r_{k'}^{\mathrm{mix}} = r^{\mathrm{mix}}(t)\big|_{t=0.25kT_x}$, where k denotes the bit index and k' is the sample index that is four times the bit index. The sequence is processed by the 1-D equalizer and the 1D-SOVA detector to generate LLRs $\lambda_{k'}$. To improve reliability, adjacent LLR samples are combined through the proposed soft-information adjustment (SIA) scheme as:

$$\tilde{\lambda}_k = \lambda_{2k'-1} + \lambda_{2k'}, \tag{1}$$

The refined LLRs are then passed to the LDPC decoder. After the initial decoding stage, turbo iterations are performed between the detector and decoder. Extrinsic information is computed as:

$$\alpha\hat{\lambda}_k^{\mathrm{ext}} = \hat{\lambda}_k - \tilde{\lambda}_k, \tag{2}$$

where $\hat{\lambda}_k$ is the decoder output LLR, and $\alpha = 0.34$ is a scaling factor used to prevent overconfident feedback. The scaled extrinsic information is fed back to the detector as a priori information for the next iteration. The best performance is achieved at 9 global iterations, as illustrated in our intrinsic study. After the final iteration, the output LLR sequence is separated into even and odd streams and threshold-detected to recover $\hat{u}_k^{\mathrm{Top}}$ and $\hat{u}_k^{\mathrm{Bot}}$.

III. Simulation Results and Conclusion

The BER performance of the proposed system is evaluated through two comparative studies, as shown in Fig. 4. In the first study, the proposed system is compared with a conventional DL-BPMR system [6]. The proposed system employs $0.25T_x$ oversampling with SIA, whereas the conventional system uses the same medium geometry with $0.5T_x$ oversampling and does not employ SIA. At a BER of 10^{-5}, the proposed system achieves an SNR gain of approximately 2 dB for both recording layers. When turbo detection with 9 global iterations is applied, additional gains of approximately 2 dB for the top layer and 3 dB for the bottom layer are obtained compared with the non-iterative case. In the second study, the proposed DL-BPMR system is compared with two single-layer BPMR systems at the same total AD of 4.0 Tb/in^2: a staggered single-layer system (SL-Staggered) and a regular-grid single-layer system (SL-Regular).

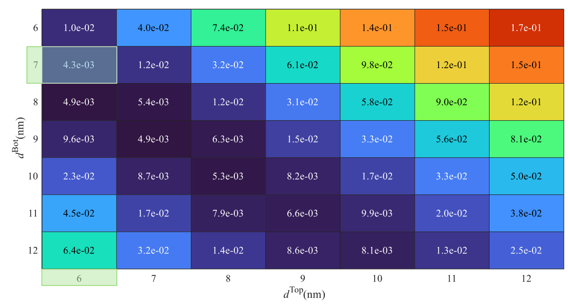

Fig. 3. Average BER performance for various top and bottom bit-island diameters at an SNR of 14 dB.

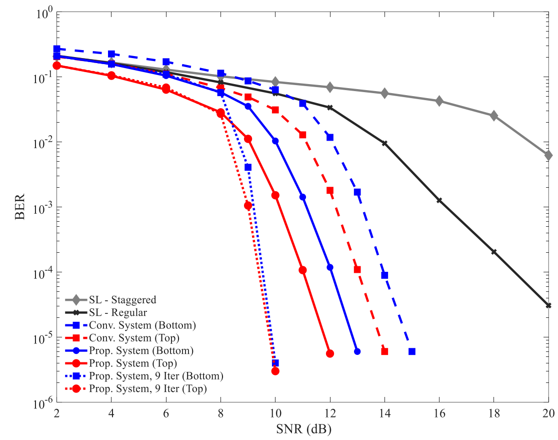

Fig. 4. BER performance comparison of the proposed, conventional DL-, and SL- BPMR systems at an AD of 4.0 Tb/in^2.

At a BER of 10^{-5}, the proposed system provides SNR gains of approximately 8 dB and 13 dB over SL-Staggered and SL-Regular, respectively. These results confirm that the proposed media-optimized iterative detection scheme significantly improves BER performance and enables efficient high-density storage for future DL-BPMR systems.

Acknowledgment

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